

Calculator Trust Test Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	calculator-trust-test

Big idea: Tools need interpretation.

Student-Facing Prompt

Inspect Calculator Trust Test Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Calculator Trust Test Stimulus A	Student-facing stimulus 1: A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Calculator Trust Test Stimulus B	Student-facing stimulus 2: A deliberately mistyped expression that creates a plausible wrong answer. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Calculator Trust Test Reveal / Twist	Student-facing stimulus 3: A trust checklist for tools: input, process, output, and independent check. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Tools need interpretation.

Actual Lesson Questions

#	Question for students
1	After inspecting Calculator Trust Test Stimulus A, what is your first knowledge claim?
2	Which exact detail from Calculator Trust Test Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Does technology make mathematical knowledge more reliable?

Ready-to-Use Examples

- A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$.
- A deliberately mistyped expression that creates a plausible wrong answer.
- A trust checklist for tools: input, process, output, and independent check.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Give examples where technology can mislead: restricted graph window, rounded decimals, regression on inappropriate data, or a graph that hides an asymptote.
2. Students explain what the tool displays and what it does not display.
3. They write a warning label for the tool.
4. Discuss responsibility for interpretation.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Does technology make mathematical knowledge more reliable?
- When can mathematical tools mislead us?
- Who is responsible for interpreting a result: tool or knower?

Knowledge Questions

- Does technology make mathematical knowledge more reliable?
- When can mathematical tools mislead us?
- Who is responsible for interpreting a result: tool or knower?

Transfer Example

For example, run this activity with a calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...